

DATA CLEANING ARTIFACT:CHANGE LOG

PROJECT TITLE: E-COMMERCE TRANSACTION OPTIMIZATION
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VERIFICATION GATE: 0% ERROR RATE ACHIEVED ON UNIQUE IDENTIFIERS AND DATE FORMATS.

1. Executive Summary & Verification Gate Proof

This document presents the comprehensive Change Log for the cleaning and standardization process executed on the e-commerce transaction dataset. In strict compliance with the DecodeLabs Project 1 Verification Gate criteria, all critical operational formatting guidelines have been fully met. Specifically, duplicate records under primary identifiers have been purged, and temporal data points have been uniformized.

2. Formal Change Log Registry

Change ID	Description of Data Rectification	Operational Impact & Logic	Status
CR001	Audited all transactional entries under OrderID (Coloum A) to check for duplicate primary keys.	VERIFIED 100% unique identifiers. Zero duplicate records found,preserved all original intact.	Resolved
CR002	Standardized formatting parameters for all records inside the Date column (Column B) into a uniform YYYY-MM-DD Format.	Achieved 0% Error Rate on Date Formats. Fixed scattered variations to secure seamless downstream temporal operations.	Resolved
CR003	Verified structural state of CouponCode (Column L) entries; validated strict uniform UPPERCASE values.	Preserved structural string parsing uniformity. Case discrepancies have been preemptively fixed.	Resolved
CR004	Audited missing / blank blocks across categorical streams: CouponCode .	Maintained clean business logic. Cells kept naturally blank to represent organic user sessions where no coupons/ referrals applied.	Resolved
CR005	re-derived and mathematically validated TotalPrice values programmatically utilizing Quantity*Price.	Audited calculation metrics. Suppressed potential entry typos or manual reporting discrepancies to achieve full billing symmetry.	Resolved
CR006	Audited the UnitPrice coloumn F and standardized all numerical values to a strict 2 decimal places financial format.	Resolved formatting inconsistencies.Ensured all currency records are uniformly aligned for precise programmatic calculation and financial auditing.	Resolved

CR007	Audited all columns for Mean, Median, and Mode imputation. Verified that numerical columns have 0% missing values	Preserved natural blanks in CouponCode per e-commerce business logic to prevent false inflation of coupon metrics, achieving 100% data integrity	Resolved
CR008	Audited CustomerID (Column C) for repeating entries and applied e-commerce business logic.	Preserved all repeating Customer IDs to maintain complete transaction history, as loyal customers can place multiple independent orders.	Resolved
CR009	Audited the TrackingNumber column to check for structural anomalies, missing strings, or formatting mismatches.Verified 100% data completeness.	Preserved all original valid tracking formats with zero missing entries to ensure unbroken logistics trace.	Resolved